

JEE MAIN REVISION MASTER: FLUID MECHANICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 12 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Fluid Statics & Dynamics (Bernoulli & Torricelli)	6 PYQs	Fluids
Moderate	Surface Tension & Viscosity (Stokes' Law)	5 PYQs	Surface & Flow
Standard	Elasticity & Hooke's Law	1 PYQs	Solids

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Fluid Statics & Dynamics (Bernoulli & Torricelli)	$A_1 v_1 = A_2 v_2$, $P + \frac{1}{2}\rho v^2 + \rho gh = \text{const}$, $v_{\text{efflux}} = \sqrt{2gh}$, $R = 2\sqrt{h(H-h)}$
Surface Tension & Viscosity (Stokes' Law)	$\Delta P_{\text{drop}} = \frac{2T}{R}$, $\Delta P_{\text{bubble}} = \frac{4T}{R}$, $h = \frac{2T \cos \theta}{\rho g r}$, $v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$
Elasticity & Hooke's Law	$Y = \frac{F/A}{\Delta L/L}$, $\Delta L = \frac{FL}{AY}$, $u = \frac{1}{2}\sigma \epsilon$, $\epsilon = \frac{\sigma^2}{2Y}$, $U = \frac{1}{2}F\Delta L$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Fluid Statics & Dynamics (Bernoulli & Torricelli) (Moderate Yield — 6 PYQs)

Core Principle: Continuity equation, Bernoulli's energy conservation principle, Torricelli's efflux theorem, and hydraulic lift.

■ **Caution / JEE Pitfall:** Speed of efflux is $\sqrt{2gh}$ where h is depth from the free liquid surface, NOT height from the ground!

■ **Speed Trick:** Maximum horizontal range of water jet occurs when hole is at mid-depth $h = H/2$, with $R_{\text{(max)}} = H$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Instrument fitted in the carburetor of the automobile to provide the correct mixture of air and fuel necessary for combustion works on..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "A horizontal pipe carries water in a streamlined flow. At a point along the pipe, where the cross-sectional area is 10cm^2 , the veloci..." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "A closed water tank has cross-sectional area A . It has a small hole at a depth of ■ from the free surface of water. The radius of the h..." [Ans: N/A]

3.2 Surface Tension & Viscosity (Stokes' Law) (Moderate Yield — 5 PYQs)

Core Principle: Surface energy, excess pressure in drops and bubbles, capillary rise formula, and terminal velocity of falling spheres.

■ **Caution / JEE Pitfall:** A soap bubble in air has TWO liquid-air interfaces, so excess pressure is $\Delta P = (4T / R)$, while a liquid drop has ONE ($\Delta P = (2T / R)$).

■ **Speed Trick:** Terminal velocity $v_t \propto r^2$. When n identical droplets coalesce: $R = n^{1/3}r$, and surface energy is released: $\Delta U = 4\pi T r^2(n - n^{2/3})$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "A 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in ..." **[Ans: C]**
- **[2025 Shift] Q#4 (MCQ):** "While determining the coefficient of viscosity of the given liquid, a spherical steel ball sinks by a distance $h=0.9\text{m}$. The radius of the ..." **[Ans: N/A]**
- **[2024 Shift] Q#1 (MCQ):** "Water flows through a horizontal pipe of varying cross-section at a rate of $0.314\text{m}^3\text{s}^{-1}$. The velocity of water at a point where the ..." **[Ans: N/A]**

3.3 Elasticity & Hooke's Law (Standard Yield — 1 PYQs)

Core Principle: Stress-strain relationships, Young's modulus, shear modulus, bulk modulus, and energy stored in stretched wires.

■ **Caution / JEE Pitfall:** Elongation due to self-weight is $\Delta L = (\rho g L^2 / 2Y) = (MgL / 2AY)$ (half of an equivalent end load Mg).

■ **Speed Trick:** When two wires are in series, equivalent Y is obtained by adding extensions: $\Delta L = \Delta L_1 + \Delta L_2$.

Representative Source PYQs:

- **[2020 Shift] Q#4 (MCQ):** "Iceberg floats in water with part of it submerged. What is the fraction of the volume of iceberg submerged, if the density of ice is ρ_{ice} ..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 30 cm long capillary tube is dipped in water, water rises upto a height of 10 cm due to capillarity. If this experiment is conducted in a freely falling water in an elevator, then the length of the water column becomes

- (A) 10 cm
- (B) 20 cm
- (C) 30 cm **[Correct]**
- (D) Zero

Correct Answer: C

Solution Analysis: In normal conditions, water rises in a capillary tube because the upward force due to surface tension balances the weight of the water column. The capillary rise is $h = (2T \cos \theta / \rho g r)$. Here, g is the acceleration due to gravity. In a freely falling elevator, the effective acceleration due to gravity becomes zero: $g_{\text{eff}} = 0$. So the water column has no effective weight. Since there is no downward weight to balance the upward pull due to surface tension, the water will rise as much as possible in the tube. But the tube is only 30 cm long. So the water can fill the whole tube.

Benchmark Problem 2: KCET 2026 (Q#2)

Instrument fitted in the carburetor of the automobile to provide the correct mixture of air and fuel necessary for combustion works on

- (A) Pascal's Law
- (B) Bernoulli's Principle
- (C) Newton's Law of Cooling
- (D) Archimede's Principle

Correct Answer: None

Solution Analysis: The correct option is: Option B: Bernoulli's Principle A carburetor mixes air and fuel in the correct ratio for combustion in an automobile engine. Inside the carburetor, air passes through a narrow region called a venturi. When air flows through this narrow part, its speed increases. According to Bernoulli's Principle, when the speed of a fluid increases, its pressure decreases. So, the pressure in the narrow region becomes less than the pressure in the fuel chamber. Due to this pressure difference, fuel is drawn into the moving air stream. This forms the air-fuel mixture needed for combustion.

Benchmark Problem 3: KCET 2020 (Q#4)

Iceberg floats in water with part of it submerged. What is the fraction of the volume of iceberg submerged, if the density of ice is $\rho_{(i)} = 0.917 \text{ g cm}^{-3}$?

- (A) 0.917
- (B) 1
- (C) 0.458
- (D) 0

Correct Answer: None

Solution Analysis: Density of ice, $\rho_{(i)} = 0.917 \text{ g cm}^{-3}$ Fraction of the volume of iceberg submerged in water is given as $\left(\frac{\text{Volume of iceberg submerged in water}}{\text{Volume of iceberg}} \right) = \left(\frac{\text{density of ice}}{\text{density of water}} \right) = \frac{0.917 \text{ g cm}^{-3}}{1 \text{ g cm}^{-3}} = 0.917$